

# Enhancing woody landscape elements

**Spatial scale:** landscape

## About this sustainable management strategy

Woody landscape elements such as hedges, tree lines, isolated trees, shrubs, and small woodland patches add structure and diversity to vineyard landscapes. Beyond defining field boundaries or providing visual character, they create a network of habitats that supports wildlife, improves ecological connectivity, and strengthens the resilience of vineyards to environmental pressures.

New plantings can be established along vineyard margins, access roads, waterways, or between vineyard blocks, while existing hedges and trees can be maintained or restored to enhance their ecological value. Selecting a mixture of native trees and shrubs with different flowering and fruiting periods provides year-round food and shelter for pollinators, birds, and other beneficial organisms. At the same time, woody vegetation helps reduce wind speed, moderates local temperatures, improves water infiltration, stabilises soils, and intercepts surface runoff.

Over time, these landscape features create a more diverse and interconnected vineyard environment. Their ecological benefits increase as vegetation matures, providing stable habitats, strengthening ecological interactions, and improving the capacity of vineyard landscapes to respond to climatic extremes.

## Key ecosystem services

- Habitat provision and biodiversity conservation (*refuge, food resources, and ecological connectivity*)
- Microclimate regulation (*wind buffering, shading, and temperature moderation*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Soil health improvement (*enhanced organic matter, nutrient cycling, and soil structure*)
- Erosion control (*root stabilisation*)
- Water retention (*reduced surface runoff*)
- Natural pest control (*support for beneficial insects and natural enemies*)
- Cultural and landscape aesthetic value (*cultural identity, structural diversity and visual character of agricultural landscapes*)

## Considerations for successful implementation

### Selection of suitable plant species

Native trees and shrubs adapted to local soil and climatic conditions generally provide greater ecological value and require less maintenance once established. A diverse mix of species with staggered flowering and fruiting periods supports other species throughout the year.

### **Strategic placement within the landscape**

The location of hedges and trees should enhance ecological connectivity while avoiding unnecessary shading of vines or interference with machinery. Existing landscape features should be retained and incorporated wherever possible.

### **Long-term establishment and maintenance**

Woody vegetation requires regular management during the establishment phase to ensure successful growth. Periodic pruning and the removal of invasive species help maintain plant health, structural stability, and ecological function over the long term.

### **Balance between ecological benefits and resource competition**

Tree and shrub species, planting density, and distance from vine rows should be selected to minimise excessive competition for water and nutrients while maximising the benefits for biodiversity, soil protection, and microclimate regulation.